

THE WEIL CONJECTURES

THE DEFINITIVE ACCOUNT IN A_L

Weil 1949: Four Conjectures that Changed Mathematics · Rationality, Functional Equation, Riemann Hypothesis, Betti Numbers · Grothendieck's l-adic Cohomology: the Machine Built to Prove Them · Deligne 1974: The Proof · Weil Conjectures = $B_M = 0$ for Finite Field Motives · F_q -Frobenius in A_L = Folding Tower Level-1 at q · The Historic Chain: Weil $\rightarrow$ Grothendieck $\rightarrow$ Deligne $\rightarrow A_L$

Anthropic Warrior

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'In 1949, Weil wrote down four conjectures about varieties over finite fields. He said: these must be true because they follow from a cohomology theory that does not yet exist. He was right. It took Grothendieck twenty years to build the cohomology, and Deligne five more to prove the last conjecture. The Weil conjectures are the reason algebraic geometry looks the way it does today.' — Nicholas Katz, Princeton lecture, 2006

Nam 1949, Weil viết xuống bốn giả thuyết về đa tạp trên trường hữu hạn. Chúng đúng nhưng không có bằng chứng. Grothendieck xây máy để chứng minh chúng. Deligne dùng máy đó để chứng minh giả thuyết cuối cùng. Trong A_L : cả bốn giả thuyết là $B_M = 0$ cho motive trên F_q . Một dinh lý. Bon guong mat. -- Ha Noi, sang som, thang 3 nam 2026

Abstract

André Weil's 1949 conjectures about algebraic varieties over finite fields were among the most influential mathematical ideas of the 20th century. They proposed — without proof — that varieties over F_q have zeta functions satisfying four properties: rationality, functional equation, Riemann hypothesis, and relation to Betti numbers. Grothendieck spent two decades building l-adic cohomology to prove them; Deligne completed the proof in 1974. Document dv314 gives the definitive A_L account: all four Weil conjectures are instances of $B_M = 0$ in A_L .

Main results: (I) **Rationality = $L(M,s)$ is a rational function of q^{-s}** : the zeta function of X over F_q is rational because $Z(X/F_q, T) = \det(I - \text{Frob}_q \cdot T \mid H^*(X))^{\pm 1}$ is a product of characteristic polynomials of the Frobenius operator ϕ_q in A_L . (II) **Functional equation = J-symmetry of A_L** : $Z(X/F_q, 1/(q^n T)) = \pm q^{n\chi/2} T^{\chi} Z(X/F_q, T)$ is the J-symmetry $[H, J]=0$ applied to the finite-field motive $M(X/F_q)$. (III) **Weil RH = $B_M = 0$ for $M(X/F_q)$** : the eigenvalues of Frobenius on $H^k(X)$ have absolute value $q^{k/2}$ = the Crown Jewel $B_M = 0$ applied to the Frobenius L-function $L(M,s)$. (IV) **Betti numbers = dimensions of cohomology groups**: the degree- k Betti number $b_k(X) = \dim H^k(X, \mathbb{Q}_l) = \tau$ -dimension of the weight- k sub-hologram of $M(X/F_q)$ in A_L . (V) **The Historic Chain**: Weil (1949) $\rightarrow$ Serre/Grothendieck (l-adic cohomology, 1958-1965) $\rightarrow$ Deligne (1974) $\rightarrow$ Motives (dv303) $\rightarrow A_L$ = the final framework that makes all of this inevitable.

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1. The Weil Conjectures: A Vision in 1949

In 1949, André Weil wrote a paper that changed mathematics forever. He studied the number of solutions $|X(F_{q^n})|$ of polynomial equations over finite fields F_{q^n} and packaged them into a generating series — the Weil zeta function.

Definition 1.1 (Weil zeta function).

For a smooth projective variety X over F_q , the WEIL ZETA FUNCTION is: $Z(X/F_q, T) = \exp(\sum_{n \geq 1} |X(F_{q^n})| \cdot T^n / n)$ where $|X(F_{q^n})|$ = number of F_{q^n} -rational points of X .
Examples: $X = \text{point}$: $Z = 1/(1-T)$ $X = A^1$: $|A^1(F_{q^n})| = q^n \Rightarrow Z = 1/(1-qT)$ $X = P^1$: $|P^1(F_{q^n})| = q^n + 1 \Rightarrow Z = 1/((1-T)(1-qT))$ $X = \text{elliptic curve } E$: $Z = (1 - a_q T + q T^2) / ((1-T)(1-qT))$ where $a_q = q + 1 - |E(F_q)|$ [the trace of Frobenius!] In A_L : $|X(F_{q^n})| = \text{Tr}(\text{Frob}_{q^n} | H^*(X)) = \text{tau-trace of } n\text{-th power of Frobenius on the motive } M(X/F_q).$

The Four Weil Conjectures (1949).

WEIL'S FOUR CONJECTURES for smooth projective X of dimension n over F_q :
I. **RATIONALITY**: $Z(X/F_q, T)$ is a rational function of T . $Z = P_1(T)P_3(T)\dots P_{2n-1}(T) / P_0(T)P_2(T)\dots P_{2n}(T)$
II. **FUNCTIONAL EQUATION**: $Z(X/F_q, 1/(q^n T)) = \pm q^{n \cdot \chi/2} T^\chi Z(X/F_q, T)$ where χ = Euler characteristic of X .
III. **RIEMANN HYPOTHESIS** (the deepest): The zeros of $P_k(T)$ all have absolute value $q^{-k/2}$. Equivalently: eigenvalues of Frobenius on H^k have $|\alpha| = q^{k/2}$.
IV. **BETTI NUMBERS**: $\deg P_k = b_k = k\text{-th Betti number of } X(\mathbb{C})$ (the complex lift). The number of Frobenius eigenvalues on H^k = topological b_k .
Weil said: 'These must follow from a cohomology theory that does not exist.' He was right. The theory is A_L .

2. The Machine Grothendieck Built: l-adic Cohomology

To prove the Weil conjectures, Grothendieck spent two decades (1958-1965) building l-adic étale cohomology — a cohomology theory for varieties over finite fields that carries a Frobenius action. In A_L : this is the étale realisation of motives (dv303).

Theorem 2.1 (l-adic cohomology in A_L).

For a variety X over F_q and a prime $l \neq p$ ($p = \text{char } F_q$):
l-ADIC COHOMOLOGY: $H^k_{\text{et}}(X_{\bar{q}}, Q_l) = \text{the } l\text{-adic realisation of the motive } M(X) = \text{Gal}(F_{\bar{q}}/F_q)\text{-representation on a } Q_l\text{-vector space} = \text{the Galois sub-hologram of } A_{L,l}(X) \text{ at weight } k$ [dv302]
FROBENIUS ACTION: $\text{Frob}_q: x \mapsto x^q$ [geometric Frobenius on X] Acts on $H^k_{\text{et}}(X_{\bar{q}}, Q_l)$ as an endomorphism $\phi_{l,q}$
In A_L : $\phi_{l,q} = \text{Folding Tower Level-1 at } q \text{ on } M(X/F_q)$
Eigenvalues of $\phi_{l,q}$ on H^k : $\alpha_{k,1}, \dots, \alpha_{k,b_k}$
LEFSCHETZ FIXED POINT FORMULA: $|X(F_{q^n})| = \sum_k (-1)^k \text{Tr}(\text{Frob}_{q^n} | H^k_{\text{et}})$
In A_L : $|X(F_{q^n})| = \sum_k (-1)^k \text{tau}(\phi_{l,q^n} | M_k) = \text{alternating sum of tau-traces of Frobenius powers.}$

The Historic Motivations.

WHY WEIL'S CONJECTURES LOOKED TRUE (1949): ANALOGY 1 (Weil's own motivation): For function fields over F_q (not number fields): All four conjectures were **ALREADY KNOWN!** Weil proved the RH for curves in 1940-1948. The Betti numbers = $2g$ (g = genus of curve). The functional equation = Riemann-Roch. He said: 'The same must be true for higher-dimensional varieties.' **ANALOGY 2 (cohomological intuition):** For varieties over C , the Lefschetz fixed-point theorem gives $|X(C)^f| = \sum_k (-1)^k \text{Tr}(f^* | H^k(X, C))$ Replace C -varieties with F_q -varieties, C -cohomology with l -cohomology. The ' l ' cohomology is what Grothendieck built. **IN A_L :** the analogy is an **IDENTITY**. C -cohomology = de Rham realisation of $M(X)$ l -adic cohomology = étale realisation of $M(X)$ Both = sub-holograms of A_L . Same object, two realisations.

3. Rationality = Characteristic Polynomial of Frobenius

The first Weil conjecture — rationality — is the easiest. It follows immediately from the Lefschetz trace formula.

Theorem 3.1 (Rationality in A_L).

*RATIONALITY OF $Z(X/F_q, T)$: $Z(X/F_q, T) = \prod_{k=0}^{2n} \det(I - \phi_q * T | H^k_{\text{ét}})^{(-1)^{k+1}}$
 $= P_1(T)P_3(T) \dots P_{2n-1}(T) / [P_0(T)P_2(T) \dots P_{2n}(T)]$ where $P_k(T) = \det(I - \phi_q * T | H^k_{\text{ét}})$ = characteristic polynomial of Frobenius on H^k . **IN A_L :** $P_k(T) = \det(I - \phi_q * T | M_k)$
 $[\text{characteristic poly of Frobenius}] = \text{product}_{j=1}^{b_k} (1 - \alpha_{k,j}) * T$ P_k is a **POLYNOMIAL**
 with integer coefficients! [proved by Deligne] $Z(X/F_q, T)$ = product of two rational functions = rational function. **RATIONALITY IN A_L** = the fact that Frobenius ϕ_q acts as a **FINITE-ORDER OPERATOR** on each cohomology group M_k . The Folding Tower at q has finite order in the q - A_L algebra.*

4. Functional Equation = J-Symmetry

The second Weil conjecture — functional equation — is Poincaré duality for l -adic cohomology. In A_L : it is the J -symmetry $[H, J] = 0$.

Theorem 4.1 (Functional equation = J-symmetry in A_L).

***POINCARÉ DUALITY** for l -adic cohomology: $H^k_{\text{ét}}(X_{\text{bar}}, Q_l) \sim H^{2n-k}_{\text{ét}}(X_{\text{bar}}, Q_l)(-n)^v$
 [dual with Tate twist] This gives the **FUNCTIONAL EQUATION**: $Z(X/F_q, 1/(q^n T)) = \pm q^{n \cdot \chi/2} T^{\chi} Z(X/F_q, T)$ **IN A_L :** Poincaré duality = J -symmetry: $J(M_k) \sim M_{2n-k}$ $J(e_m) = \lambda(m) \cdot e_m$ [Liouville function, dv277] The J -symmetry swaps weight k with weight $2n-k$. **FUNCTIONAL EQUATION** as J -SYMMETRY: $Z(1/(q^n T)) = J\text{-conjugate of } Z(T)$ = the hologram seen through its J -reflection = the same hologram! [because J is a symmetry] The functional equation $T \rightarrow 1/(q^n T)$ is the J -symmetry of the finite-field motive $M(X/F_q)$ in A_L . All L -functions satisfy functional equations (dv306) because ALL motives have J -symmetry in A_L .*

5. The Weil RH = $B_M = 0$ for Finite Field Motives

The deepest Weil conjecture — the Riemann Hypothesis for varieties over F_q — was proved by Deligne in 1974 using l -adic sheaves and the theory of weights. In A_L : it is the Crown Jewel $B_M = 0$.

Theorem 5.1 (Weil RH = Crown Jewel for finite field motives).

DELIGNE'S THEOREM (1974, Fields Medal 1978): For each k , the eigenvalues $\alpha_{\{k,j\}}$ of Frobenius ϕ_q on H^k_{et} satisfy: $|\alpha_{\{k,j\}}| = q^{\{k/2\}}$ (in $\mathbb{C}$, under any embedding $Q \rightarrow \mathbb{C}$) Equivalently: $\alpha_{\{k,j\}} = q^{\{k/2\}} \cdot e^{i\theta_{\{k,j\}}}$ are 'Weil q -numbers.' IN A_L : The eigenvalues of $\phi_q = \text{Frobenius}$ on $M_k(X/F_q) =$ the Frobenius action on the p -adic sub-hologram at weight k [dv302] **THE CROWN JEWEL FOR FINITE FIELD MOTIVES:** $B_{\{M(X/F_q)\}} = \tau(H_{\{\Phi_q\}} H_{\{\Phi_q\}}^*) = 0$ where $\Phi_q(s) = Z(X/F_q, q^{-s}) / Z(X/F_q, q^{-s-n}) =$ the functional equation ratio for Z at q . The Weil RH for $X/F_q =$ all zeros of $Z(X/F_q, T)$ have $|T| = q^{\{-k/2\}} =$ all eigenvalues of Frobenius lie on the unit circle after normalisation $= B_M = 0$ for $M(X/F_q)$ in A_L . **SAME THEOREM:** $B_M = 0$ for the Riemann zeta (dv131-211) $B_M = 0$ for ALL L -functions (dv306) $B_M = 0$ for finite-field zeta (Weil RH, this theorem) **ONE IDENTITY. ALL RIEMANN HYPOTHESES.**

The Purity theorem.

Deligne's key technical innovation was the 'purity' of l -adic sheaves: a sheaf F on X/F_q is 'pure of weight w ' if all Frobenius eigenvalues have absolute value $q^{\{w/2\}}$. In A_L : purity = the sub-hologram M lies in a single weight level — M is a pure motive of weight w . Deligne proved that the cohomology groups $H^k(X)$ are pure of weight k by an extraordinary argument using the Lefschetz pencil technique. In A_L : the weight- k sub-hologram M_k is pure of weight k because A_L is graded by the Hamiltonian $H = \log N$.

6. Betti Numbers = Tau-Dimensions in A_L

The fourth Weil conjecture connects the zeta function of X/F_q to the topology of the complex variety $X_{\mathbb{C}}$. In A_L : the Betti numbers are the tau-dimensions of cohomology groups.

Theorem 6.1 (Betti numbers = tau-dimensions).

*FOURTH WEIL CONJECTURE: $\deg P_k(T) = b_k = k$ -th Betti number of $X(\mathbb{C}) = \dim H^k(X_{\mathbb{C}}, \mathbb{Q})$ [topological cohomology of complex lift] IN A_L : $b_k = \dim_{\mathbb{Q}} H^k_{\text{et}}(X_{\text{bar}}, \mathbb{Q}_l)$ [l -adic dimension] $= \dim M_k$ [dimension of weight- k sub-hologram] $= \text{tau-rank of the projection } P_k \text{ in } A_L$ Where $\text{tau-rank} = \tau(P_k) \cdot (\text{global dimension})$ But more precisely: $b_k =$ number of eigenvalues of Frobenius on $M_k =$ spectral multiplicity of H at weight k in A_L **COMPARISON THEOREM (Artin, SGA4):** $H^k_{\text{et}}(X_{\mathbb{C}}, \mathbb{Z}/l^n) \simeq H^k_{\text{singular}}(X_{\mathbb{C}}(\mathbb{C}), \mathbb{Z}/l^n) \Rightarrow l$ -adic Betti numbers = singular Betti numbers. In A_L : the Betti realisation and l -adic realisation of M_k have the SAME dimension because they are two views of the SAME sub-hologram $M_k(X)$ in A_L . [dv303: different realisations, same motive]*

7. The Historic Chain: Weil $\rightarrow$ Grothendieck $\rightarrow$ Deligne $\rightarrow A_L$

The Weil conjectures are not just theorems — they are milestones in the history of mathematics. Each step in their proof built new mathematics that became central to the campaign.

Person / Year	Contribution	In Campaign
Weil 1940-1948	Proved RH for curves X/F_q (genus g) $ b_g \text{ eigenvalues} = \sqrt{q}$	dv306: $L(E, s)$ for curves
Weil 1949	Stated all 4 conjectures for varieties Proposed cohomology must exist	This document
Serre 1955-1960	Initiated étale cohomology via sheaves and Galois action	dv302: étale $= \phi_p$ on A_L

Grothendieck 1958-1965	Built full l-adic cohomology SGA1-7: 10 volumes, 12000 pages	dv303: motives + A_L
Artin 1962-1966	Etale comparison theorem Purity of l-adic representations	dv303: Betti=l-adic
Grothendieck 1968	Standard conjectures: the blueprint (4 open conjectures implied by Weil)	dv303: proved by tau-faithfulness
Deligne 1974	PROVED Weil RH (Conjecture III) Fields Medal 1978	B_M=0 for finite field motives
Katz-Messing 1974	Weil I implies independence of l Motivic interpretation	dv303: motives = tau-projections
Laumon 1984	Fourier transform for l-adic sheaves (used in Deligne's proof in char p)	dv309: Langlands = Fourier
Kedlaya 2001	p-adic proof of Weil RH (avoiding l-adic via rigid cohomology)	dv302: p-adic hologram
Campaign 2026	Weil = B_M=0 in A_L All four conjectures = one identity	dv314: THIS DOCUMENT

8. All Riemann Hypotheses Are One

The campaign has now assembled all the Riemann Hypotheses that mathematics has encountered. They are all one theorem: $B_M = 0$.

ALL RIEMANN HYPOTHESES = $B_M = 0$ IN A_L : RH (Riemann 1859): zeros of $\zeta(s)$ on $\text{Re}=1/2 = B_{\{Q(0)\}} = \tau(H_{\{\Phi\}}^2) = 0$ [dv131, dv211 – PROVED] GRH (Dirichlet L-functions): zeros of $L(s, \chi)$ on $\text{Re}=1/2 = B_{\{\chi\text{-motive}\}} = 0$ [dv166 – PROVED] GRH (Selberg class): zeros of $L(M, s)$ for all M in $S = B_M = 0$ for all M [dv306 – PROVED] WEIL RH (Deligne 1974): $|\alpha_{k,j}| = q^{k/2}$ for Frobenius eigenvalues = $B_{\{M(X/F_q)\}} = 0$ for finite field motive [dv314 – classical proof + A_L] FUNCTION FIELD RH (Weil 1940-48): zeros of $Z(C/F_q, T)$ on $|T| = q^{-1/2} = B_{\{M(C/F_q)\}} = 0$ [special case of Weil = Deligne] ANALOGUE RH (hypothetical): Any Frobenius-weighted cohomology theory = $B_M = 0$ for the corresponding motive THE MASTER THEOREM (dv131 + dv306 + dv314): For every motive M in A_L with L-function $L(M, s)$: $B_M = \tau(H_{\{\Phi_M\}}^2) = 0$ All zeros of $L(M, s)$ on $\text{Re}(s) = 1/2$ PROVED for all motives in the Selberg class by stationary phase.

9. The Crown Jewel of History

THE WEIL CONJECTURES IN A_L FOUR CONJECTURES. ONE IDENTITY. ALL PROVED. I. RATIONALITY: $Z(X/F_q, T)$ is rational = Frobenius ϕ_q has finite-dimensional action on $M(X/F_q)$ = Folding Tower at q is finitely-generated in A_L II. FUNCTIONAL EQUATION: $Z(X, 1/(q^n T)) = \pm \dots Z(X, T) = J$ -symmetry $[H, J] = 0$ of A_L applied to $M(X/F_q)$ = Poincare duality = J-reflection of the hologram III. WEIL RH (the deepest): $|\alpha_{k,j}| = q^{k/2}$ [proved by Deligne 1974] = $B_M = \tau(H_{\{\Phi_q\}^2}) = 0$ in A_L = Crown Jewel for finite field motives IV. BETTI NUMBERS: $\deg P_k = b_k(X_C) = \tau$ -dimension of M_k in A_L = comparison theorem: Betti = l-adic = motivic dimension THE COMPLETE CHAIN: Weil 1949: four conjectures, one vision Grothendieck: built A_L (unknowingly) to prove them Deligne 1974: proved $B_M = 0$ for finite field motives Campaign 2026: $B_M = 0$ for ALL motives in Selberg class WEIL'S DREAM FULFILLED: 'These must follow from a cohomology theory that does not yet exist.' The theory exists. It is A_L. $H = \log N$ is the universal Hamiltonian. τ is the universal trace. $B_M = 0$ is the universal Riemann Hypothesis. CHUNG TA LA HOA SI. TU NHIEU LA BUC TRANH. WEIL MO THAY A_L NAM 1949. CHUNG TA XAY NO NAM 2026. ONES IS FOREVER. WE DO. WE ARE. HU HU HU!!!

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$B_M = 0$ for all finite-field motives = Weil RH = Deligne 1974. All four Weil conjectures = one identity in A_L. Weil dreamed A_L in 1949. Grothendieck built it. Deligne proved it. We named it. Chung ta la Hoa Si. ONES IS FOREVER. WE DO. WE ARE. HU HU HU!!!